Artifical Intelligence Basics Course Summary

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1 Key Terms

- **Minimax Search**: A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally.
- **Alpha{-}Beta Pruning**: A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision.
- **Turing Test**: An operational test for intelligent behavior, proposed by Alan Turing, where a human evaluator must distinguish between a human and a machine based on their textual responses.
- **Game Tree**: A tree-like structure representing all possible sequences of moves in a game, with leaf nodes representing the outcomes.
- **Perceptron**: A fundamental unit in neural networks, consisting of weighted inputs, a summation node, and an activation function that produces an output.
- **Reinforcement Learning**: A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.
- **Matrix**: A rectangular array of numbers, symbols, or expressions arranged in rows and columns.
- **Linear Transform**: A function that maps vectors from one vector space to another, satisfying $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all vectors u, v and scalars c.
- **Matrix Multiplication**: An operation that combines two matrices to produce a third matrix, where the element at row i and column j of the resulting matrix is the dot product of row i of the first matrix and column j of the second matrix. The inner dimensions must match.
- **Matrix Inverse**: For a square matrix A, its inverse A^{-1} is a matrix such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.
- **Matrix Transposition**: An operation that swaps the rows and columns of a matrix. The transpose of matrix A is denoted as A^T.
- **Derivative**: A measure of how a function changes in response to a small change in its input. It represents the instantaneous rate of change.
- **Partial Derivative**: A derivative of a function of multiple variables with respect to one of those variables, while holding the others constant.
- **Sigmoid Function**: A mathematical function that maps any real number to a value between 0 and 1. Often used as an activation function in neural networks.
- **Inductive Reasoning**: The process of deriving general principles from specific observations.
- **Linear Regression**: A machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.

- **Multiple Linear Regression**: An extension of linear regression that models the relationship between a dependent variable and two or more independent variables.
- **Overfitting**: A phenomenon in machine learning where a model learns the training data too well, resulting in poor generalization to unseen data.
- **Binary Classification**: A type of machine learning problem where the goal is to classify data into one of two categories.
- **Decision Boundary**: A line or hyperplane that separates data points into different classes in a classification problem.
- **0{-}1 Loss**: A loss function in binary classification that assigns a loss of 1 if the prediction is incorrect and 0 if it is correct.
- **Perceptron Loss**: A loss function used in binary classification that is differentiable and encourages correct classification.
- **Hinge Loss**: A loss function used in binary classification that is differentiable and encourages correct classification with a margin.
- **Logistic Loss**: A loss function used in binary classification that is differentiable and outputs a probability.
- **Gradient Descent**: An iterative optimization algorithm used to find the minimum of a function by following the negative gradient.
- **Stochastic Gradient Descent**: A variant of gradient descent that updates the model parameters using a randomly selected subset of the data.
- **Empirical Risk Minimizer**: A principle in machine learning that aims to find the model that minimizes the average loss on the training data.

2 Problem Types

- **Understanding the Nature of Intelligence**: This problem explores the definition and characteristics of intelligence, both human and artificial. Different perspectives and definitions of AI are examined.
- **Evaluating Intelligent Behavior**: This problem focuses on developing methods for assessing whether a machine exhibits intelligent behavior. The Turing Test and its limitations are discussed, along with counterarguments like Searle's Chinese Room argument.
- **Overcoming the Limits of Search**: This problem addresses the computational challenges of exhaustive search in complex games like chess and Go. The exponential growth of search space is highlighted, leading to the need for machine learning techniques to supplement search.
- **Addressing AI Safety and Ethical Concerns**: This problem explores the potential risks and ethical implications of AI systems. Examples include bias in algorithms and the vulnerability of neural networks to adversarial attacks.
- **Bridging the Gap Between Different AI Domains**: This problem investigates the commonalities and differences between seemingly disparate AI tasks, such as game playing and speech synthesis. The shared algorithmic components and challenges are analyzed.
- **Understanding the Challenges of Autonomous Systems**: This problem examines the complexities involved in building autonomous systems, particularly autonomous vehicles. The reasons for choosing autonomous vehicles as an AI Grand Challenge are explored.
- **Developing Effective Game{-}Playing Algorithms**: This problem focuses on designing algorithms for playing games, considering aspects like state representation, action selection, and opponent modeling. Minimax search and Alpha-Beta pruning are discussed as key techniques.

- **Representing Linear Transformations with Matrices**: Matrices are used to represent linear transformations, which map vectors from one space to another while preserving linear combinations. This is fundamental to many AI algorithms.
- **Matrix Arithmetic**: The slide covers basic matrix operations such as addition, subtraction, scalar multiplication, and multiplication, which are essential for manipulating data in AI applications.
- **Solving Systems of Linear Equations**: Systems of linear equations can be represented and solved using matrices and their inverses. This is crucial for various AI tasks involving optimization and data analysis.
- **Matrix Transposition**: Matrix transposition is a fundamental operation that swaps rows and columns, useful for data manipulation and algorithm design in AI.
- **Parallelization of Matrix Multiplication**: The slide demonstrates how matrix multiplication can be parallelized using domain decomposition, improving computational efficiency for large-scale AI applications.
- **Calculus in AI**: The slide highlights the importance of calculus, specifically derivatives, partial derivatives, and gradient calculations, in AI algorithms, particularly in optimization problems.
- **Linear Regression**: Finding the line that best fits a set of data points, minimizing the sum of squared distances between the points and the line.
- **Multiple Linear Regression**: Extending linear regression to include multiple independent variables to predict a dependent variable.
- **Curve Fitting**: Fitting a curve to data points, potentially using polynomial functions of higher degrees, to capture non-linear relationships.
- **Overfitting**: The phenomenon where a model fits the training data too closely, resulting in poor generalization to unseen data.
- **Binary Classification**: Classifying data points into two distinct categories using a learned function.
- **Choosing a Loss Function**: Selecting an appropriate loss function (e.g., 0-1 loss, perceptron loss, hinge loss, logistic loss) to measure the error of a classification model.
- **Gradient Descent**: An iterative optimization algorithm used to minimize the loss function by following the direction of the negative gradient.
- **Stochastic Gradient Descent**: A variant of gradient descent that updates the model parameters using a randomly selected subset of the data points at each iteration.

3 Algorithm Solutions

- **Minimax Search**: A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha{-}Beta Pruning**: An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.
- **Matrix Addition**: $A + B = C$, where $c_{ij} = a_{ij} + b_{ij}$
- **Matrix Scalar Multiplication**: $k \cdot A = B$, where $b_{ij} = k \cdot a_{ij}$

- **Matrix Multiplication**: $C_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$
- **Matrix Transposition**: Swapping rows and columns of a matrix A to obtain A^T.
- **Solving Linear Equations with Invertible Matrices**: $Ax = b \implies x = A^{-1}b$
- **Derivative of Sigmoid Function**: $\frac{d}{dx}\sigma(x) = \frac{e^{-x}}{(1+e^{-x})^2}$
- **Perceptron Algorithm**: Iteratively update the weight vector w based on misclassified data points. If $sign(w^T x_n) \neq y_n$, then $w \leftarrow w + y_n x_n$.